AUTOMATED PROBLEM SOLVING BY SEARCH

- Generalized Techniques for Solving Large Classes of Complex Problems
- Problem Statement is the Input and solution is the Output, sometimes even the problem specific algorithm or method could be the Output
- Problem Formulation by AI Search Methods consists of the following key concepts
  - Configuration or State
  - Constraints or Definitions of Valid Configurations / Rules
  - Rules for Change of State and their Outcomes → State Changes / Algorithm.
  - Initial or Start Configurations
  - Goal Satisfying Configurations
  - An Implicit State or Configuration Space → State Space
  - Valid Solutions from Start to Goal in the State Space
  - General Algorithms which SEARCH for Solutions in this State Space
- ISSUES (AI).
  - Size of the Implicit Space, Capturing Domain Knowledge, Intelligent Algorithms that work in reasonable time and Memory, Handling Incompleteness and Uncertainty

TOWER OF HANOI

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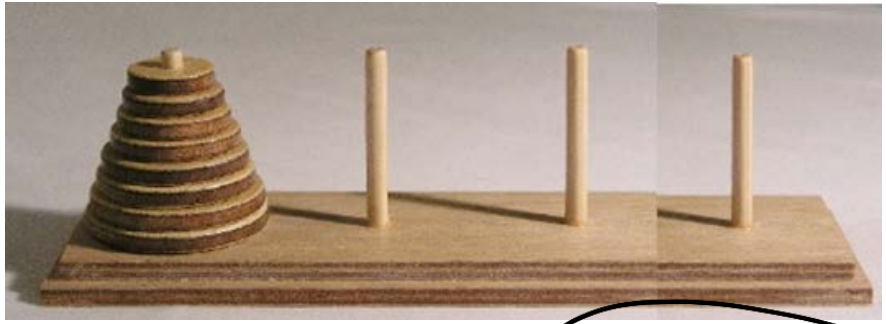
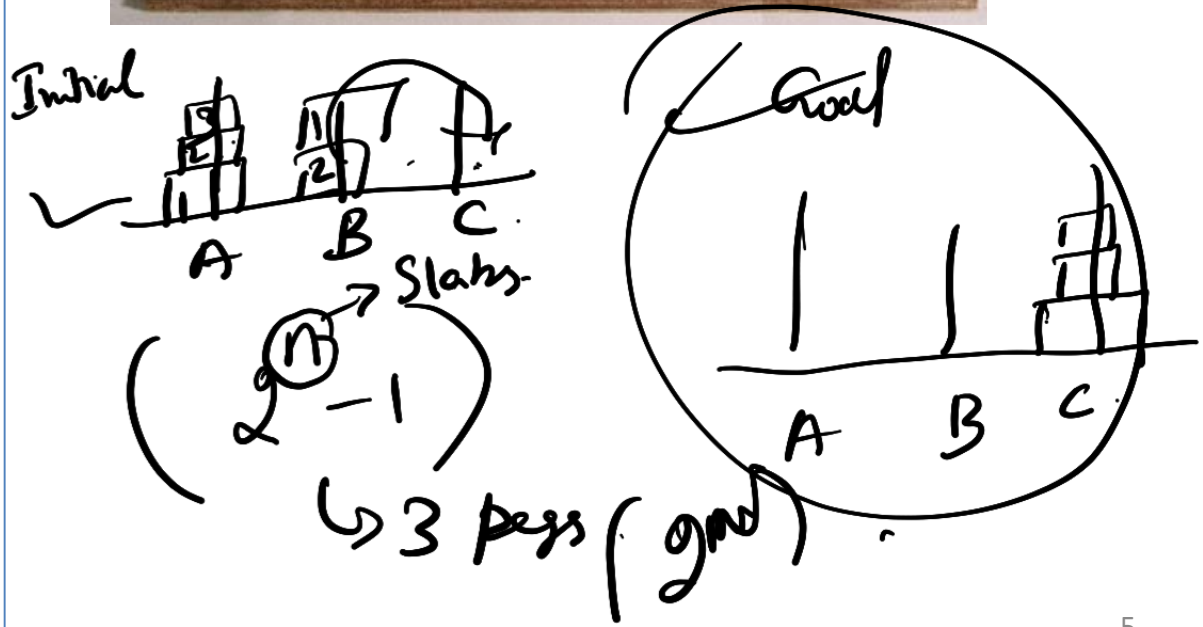
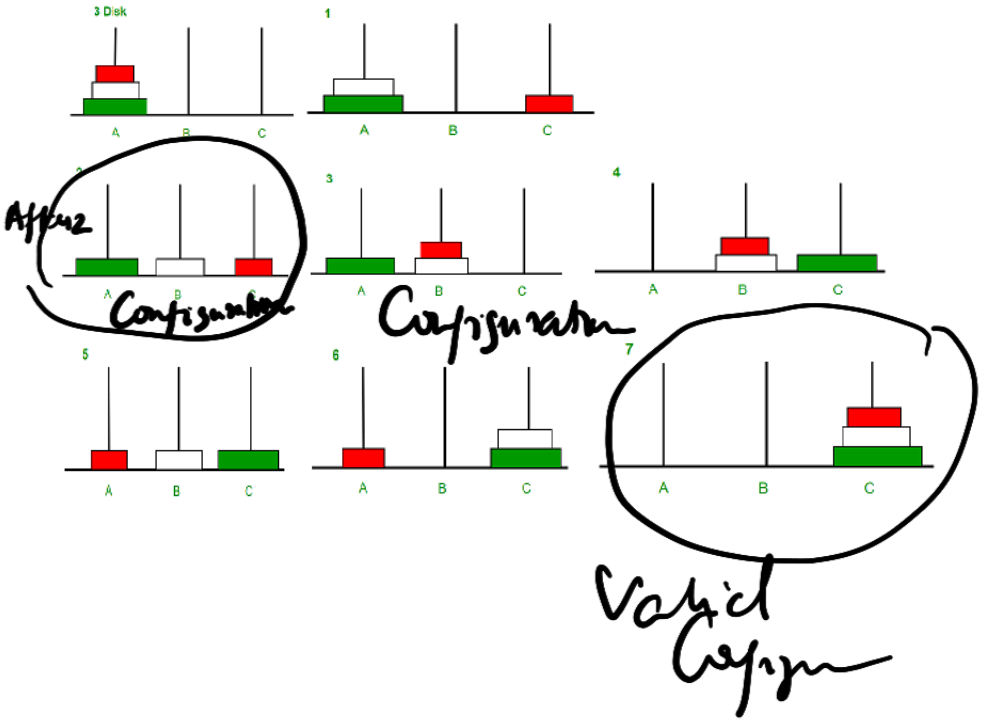
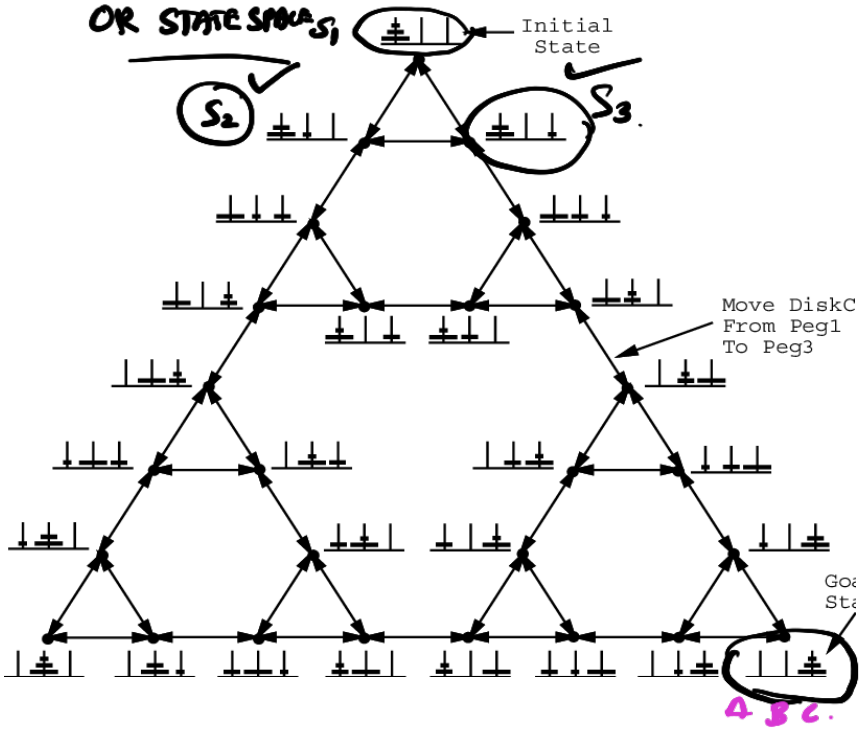




3 DISK, 3 PEG TOWER of HANOI STATE SPACE





Search :-

process of looking for a sequence of actions to reach a goal state.

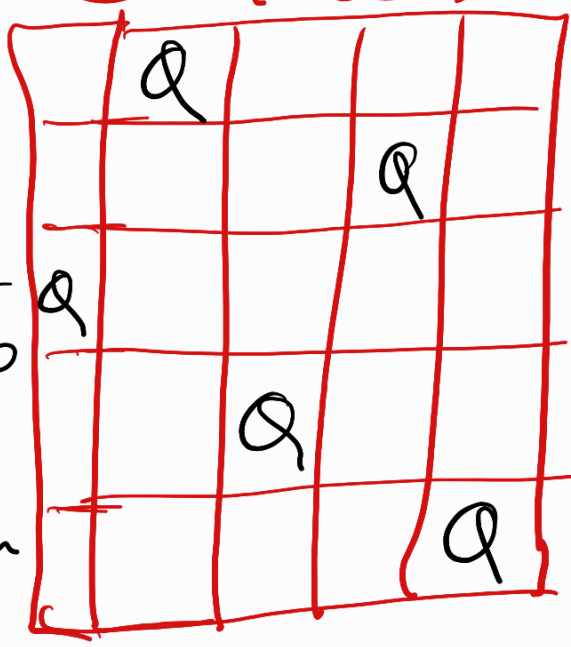
Search Algorithm :-

- Given an input ← Problem
- Expected to find a o/ps ← Solution
- To identify the sequence of actions to reach the goal State.
- If a solution is met that corresponds to the goal state, then it's task is executed.
  - Execution State.

Well-defined Problem:-

5 Components

5-Queens



① Initial State :- An empty board that can hold 0 to 5 Queens.

② Actions :- Add the queen one by one in the boards.

③ Goal test :- All the 5 Queens must be placed in the board.

④ Transition model :- Valid Configuration of states must be ensured at each sequence of placing the queen

⑤ Path Cost , States